

Data Network Design and Management 1

CNS 2 SEMESTER 1

UNIT CODE: CNS 2104



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Module leader/Class Rep: TBD

Staffroom: HP Lab, STC Floor 1

Phone No: (Classreps Prioritized)

Meeting Hours: As per timetable

AIM OF THE UNIT

- Appreciate the infrastructure that supports communication between networks, devices and users.
- How do we conceptualize, design and deploy this infrastructure? How do you manage it?

IMPORTANT INFORMATION

- What are the pros and cons of this unit?
- Learning outcomes
- Plagiarism (Q&A will reveal this)
- Build relationships (Optional)

SKILLS AND GAINS

- Cisco Packet Tracer
 - All Labs can be simulated on the tool.
 - Simulations reduce error rates on actual environments
- Actual Network Devices
 - Configure and interconnect switches and router to illustrate enterprise networks
- Network Function Virtualization
- Teamwork
 - You're expected to work as individuals but teamwork and consultations are highly encouraged to understand team dynamics in working environment.

COURSE DELIVERY

- Online Sessions(Seminars on Writing)
 - A- Thursdays 13:15-15:15
 - B- Friday 10:15-12:15
- Physical Class (Seminars, Assessments)
 - A- Wednesdays 08:15-10:15
 - B- Fridays 08:15-10:15am

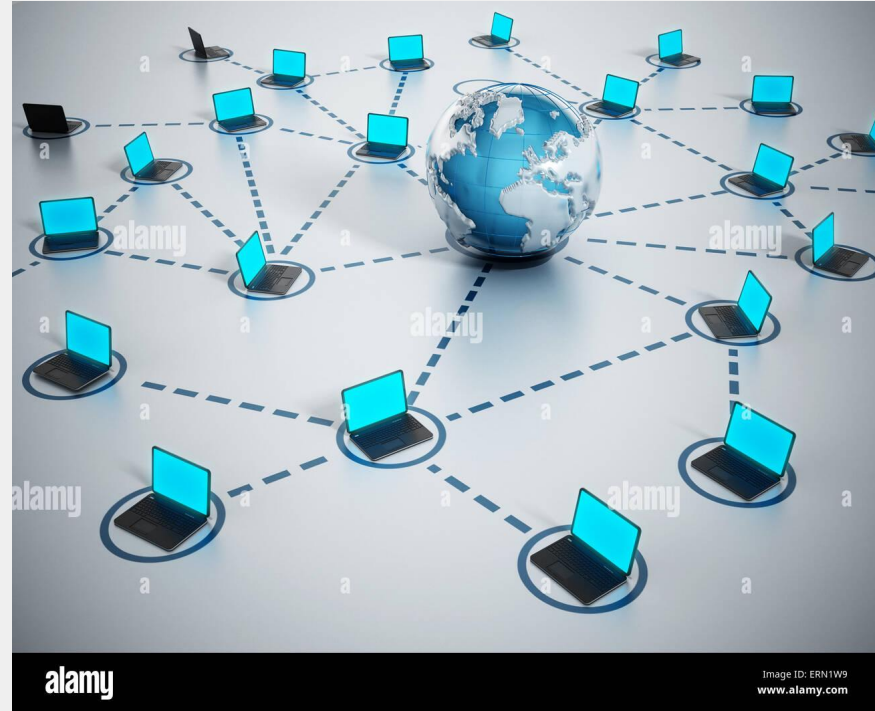
COURSE ASSESSMENT BREAKDOWN

Consider the following:

- Coursework- 50%
 - CAT 1- 10%- Week 4-5
 - CAT 2- 10%- Week 8-9
 - CAT 3*
 - Group Presentations- 10%
 - Lab Assignments- 10%
- Final Exam- 50%

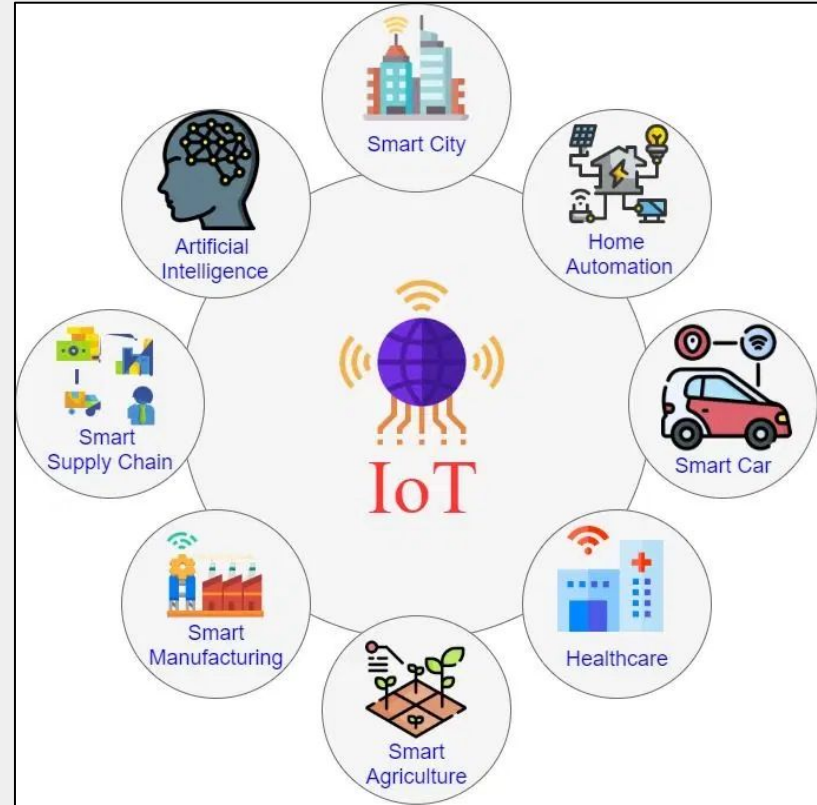
Why DNDM

- **The backbone of modern communication**—how data travels from your device to servers across the globe.
- **Key design principles**—devices, topologies, protocols, and scalability to handle billions of users-
- **Management strategies**—monitoring performance, optimizing traffic, and defending against cyber threats.



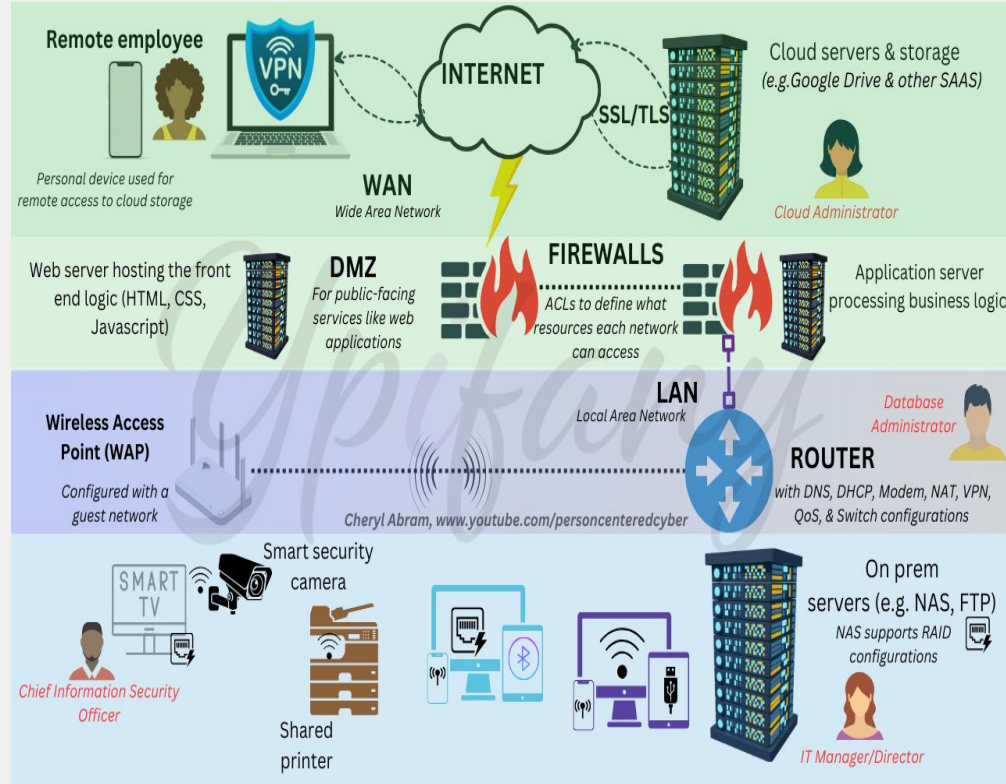
Why Networks Matter

- Without networks—no internet, no cloud, no instant messaging.
- Every click, every transaction, every video call depends on well-designed networks."



Why Design

- The blueprint of how devices connect and communicate.
- Key Elements:
 - Topologies (Star, Mesh, Ring).
 - Protocols (TCP/IP, HTTP, FTP).
 - Physical Links
 - Logical links
 - Hardware (Routers, Switches, Cables)
 - Software (OSs, Hypervisors)



Why Manage

- Design is just the start—management keeps networks alive.
- Variables
 - Scalability – Handling growth (IoT, 5G, AI).
 - Performance – Minimizing latency, maximizing speed.
 - Security – Defending against cyber threats.
 - Cost Efficiency – Balancing budget & performance.



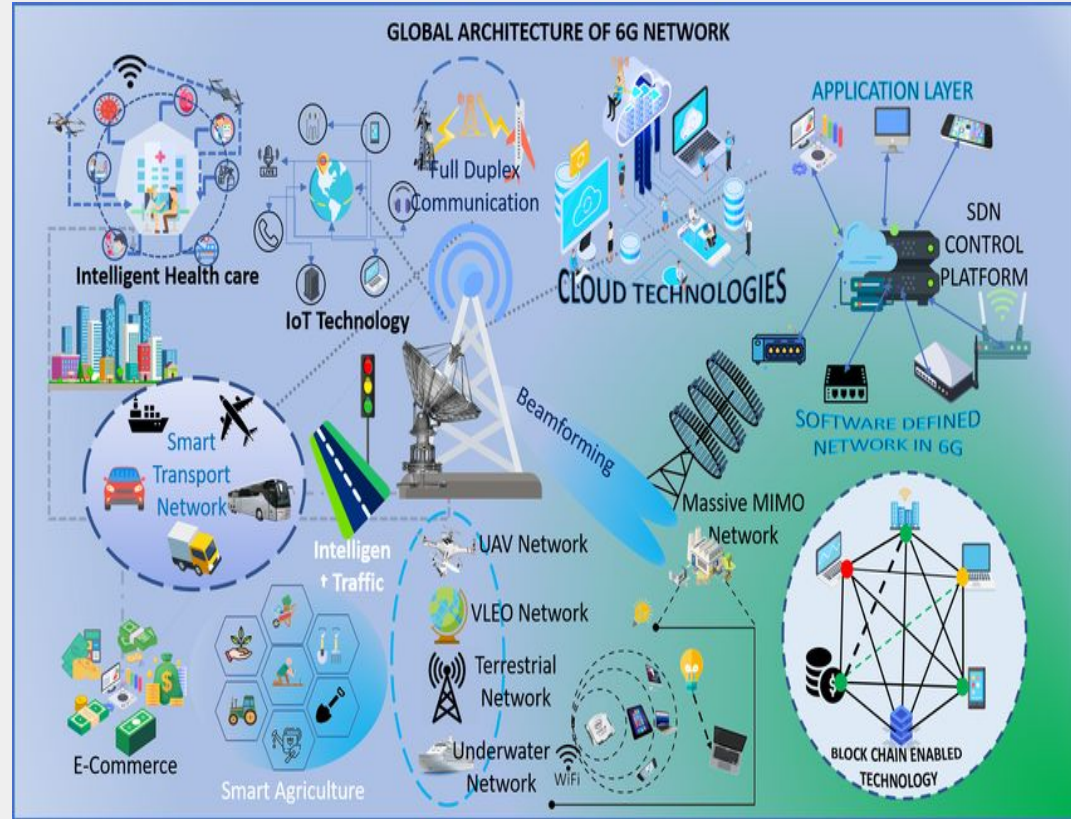
Key Tasks:

- Monitoring traffic & performance.
- Troubleshooting failures.
- Upgrading infrastructure.
- Enforcing security policies

What Next

- Deep dive into protocols & architectures.
- Design for services and optimal data transfer
- Hands-on network simulations (e.g., Cisco Packet Tracer, GNS3).
- Cybersecurity best practices.

Networks are like the nervous system of the digital age—design them well, and the world runs smoothly.



Classroom and Consultation Time

- LMS Platform- Google Classroom
- Classroom code: zeuaiph6

Link:

[https://classroom.google.com/c/NzY1MjI5NTQ0Njgx?cjc=zeuai
ph6](https://classroom.google.com/c/NzY1MjI5NTQ0Njgx?cjc=zeuai
ph6)

- Group A- Wednesdays 10:15am-12:15pm
- Group B: Fridays 12:15-2:15pm

CLASS RULES

- **Punctuality** - Always appreciated but this is, up to you, do the right thing. (Sign the Register)
- Class Participation: Presentations and discussions on reading assignments and activities is highly encouraged
- Peer Learning and positive criticism from each other is encouraged-
 - Remember neurodivergent and neurotypical perspectives

IMPORTANT TAKEAWAYS

- Compound Effect applies to life. **ALWAYS**
 - Daily practice makes a huge difference than working towards the goal overnight
- Focus on what you want to achieve by the end of your degree.
 - Always remember the bigger picture
- As everyone says “It’s tough to accept, no one cares. 😊”
 - People can only care as much as you care for yourself
 - Care for self and classmates- perspectives grow insights

To Do List

- Enroll to Google Classroom
- Review the Outline and reference texts
- Coursework Weighting
- Attempts all Assignments and Lab exercises